

```
#include <stdio.h>
```

```
#define SIZE 10
```

```
int main()
```

```
{
```

```
    int id[] = {105,210,315,420,525,630,735,840};
```

```
    int n = 8, h[SIZE], i, j, pos, col = 0;
```

```
    int key, hashComp = 0, linearComp = 0;
```

```
    for(i = 0; i < SIZE; i++)
```

```
        h[i] = -1;
```

```
    printf("HASH TABLE INSERTION\n");
```

```
    for(i = 0; i < n; i++)
```

```
    {
```

```
        pos = id[i] % SIZE;
```

```
        while(h[pos] != -1)
```

```
        {
```

```
            col++;
```

```
            pos = (pos + 1) % SIZE;
```

```
        }
```

```
        h[pos] = id[i];
```

```
printf("\nAfter inserting %d:\n", id[i]);
```

```
for(j = 0; j < SIZE; j++)
```

```
    printf("%d : %d\n", j, h[j]);
```

```
}
```

```
printf("\nTotal collisions = %d\n", col);
```

```
/* Search using Hashing */
```

```
printf("\nEnter ID to search: ");
```

```
scanf("%d", &key);
```

```
pos = key % SIZE;
```

```
for(i = 0; i < SIZE; i++)
```

```
{
```

```
    hashComp++;
```

```
    if(h[pos] == key)
```

```
{
```

```
    printf("Hashing: Found at index %d\n", pos);
```

```
    break;
```

```
}
```

```
if(h[pos] == -1)
```

```
    break;
```

```
    pos = (pos + 1) % SIZE;
}

/* Linear Search */
for(i = 0; i < n; i++)
{
    linearComp++;

    if(id[i] == key)
    {
        printf("Linear Search: Found at position %d\n", i);
        break;
    }
}

printf("\nHashing comparisons = %d\n", hashComp);
printf("Linear search comparisons = %d\n", linearComp);

printf("\nLoad Factor = %.2f\n", (float)n / SIZE);

return 0;
}
```